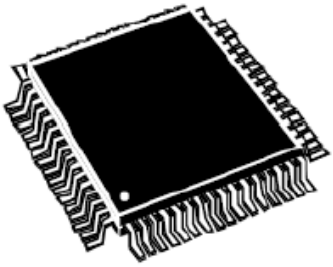
Tutorial 2 - An Overview of STM32F103

Q2.1 Which of the following is an actuator device?

(a) Infra-Red (IR) sensor
(b) Servo Motor
(c) Light Dependent Resistor (LDR)
(d) Push button

b

Q2.2 Which package is used for STM32F103RB for the NUCLEO F103RB development board?



C

(a) VFQFPN
(b) UFQFPN
(c) LQFP
(d) BGA

Q2.3 In Mbed, what is the purpose of the Linker in the build toolchain?

(a) To generate object files from the C source code files after compilation
(b) To link all the C code files into a single binary hex file
(c) To generate Assembly files from the compiled C source code files
(d) To link all the Object files into a single binary executable file

d

Q2.4 The Max CPU frequency on the NUCLEO F103RB development board is _____.

(a) 72MHz
(b) 64MHz
(c) 48MHz
(d) 100MHz

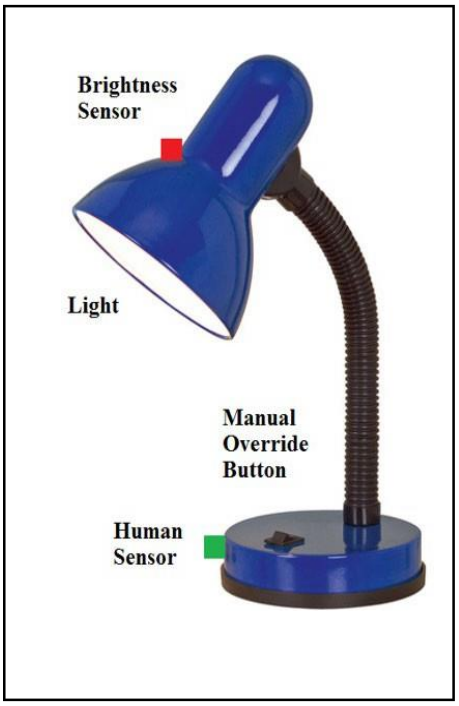
a

Q2.5 To run at the Max CPU frequency, the NUCLEO F103RB development board is using _____.

(a) HSE @ 8MHz
(b) HSI @ 8MHz
(c) LSE @ 32kHz
(d) LSI @ 40kHz

a

Q2.6 The diagram below shows the “Smart Reading Lamp”.



The STM32F103RB microcontroller is used for the implementation of an “intelligent reading lamp”.

If it is dark (Brightness Sensor, B == 0) and human is detected (Human Sensor, H == 1), then the light will be switched on automatically (Light, L = 1).

The light will be off if it is bright (B == 1) or human is not detected (H == 0).

Manual override Button, M, is provided so that if M == 1, then the light will be turned on, regardless of the status of the Brightness Sensor or Human Sensor.

In other words, the light will only depend on the sensors’ status when M == 0.

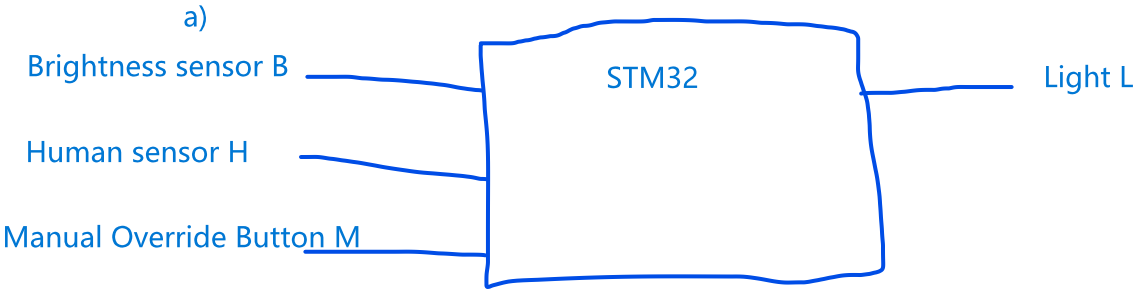
(a) Draw the system block diagram to show how the Brightness Sensor B, the Human Sensor H, the Manual Override Button M and the Light L can be connected the STM32 microcontroller.

(b) Draw the flowchart for the intelligent control of the “Smart Reading Lamp”

(c) Which type of sensors could be used for the Brightness Sensor and the Human Sensor?

(d) Explain why the brightness sensor mounted above the lamp shade.

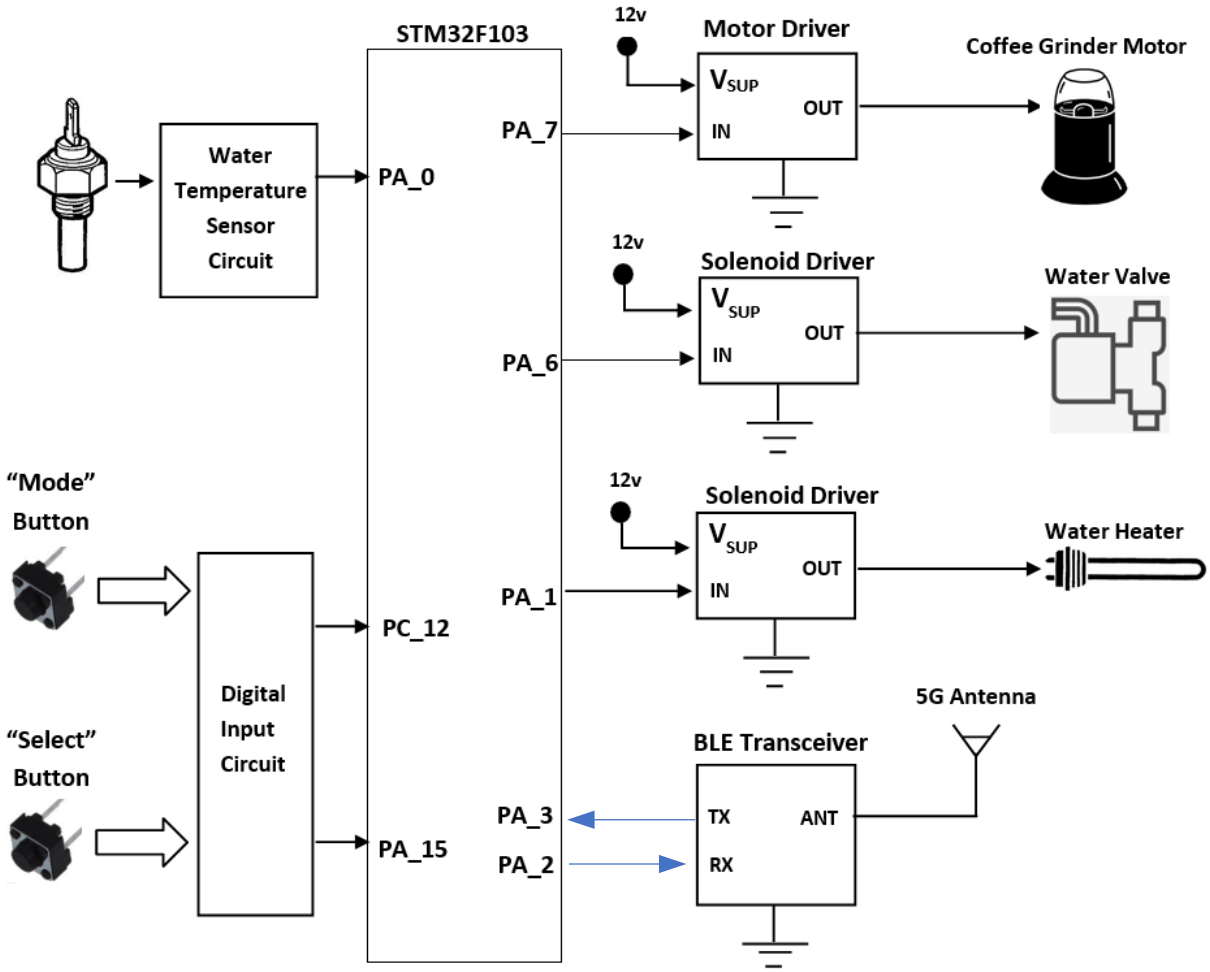
a)



c) Thermistor/IR sensor, Ultrasonic sensor.

d) Brightness sensor is mounted above the lamp shade in order to prevent a false reading from the sensor due to the result of the lamp's emitted light.

Q2.5 The diagram below shows the System Architecture of a Smart IoT coffee machine.



To make a cup of coffee, the water first needs to be boiled to a target temperature of 100 °C. Once the target water temperature is reached, the water heater needs to be turned off.

Next the coffee beans need to be ground to coffee powder by activating the grinder motor for 10 seconds.

The grinder is then deactivated which then automatically dispenses the ground coffee powder into an internal mixing chamber with the boiled water.

The coffee drink is then finally dispensed into a cup and a BLE (Bluetooth Low Energy) message is sent to a mobile phone to inform the user the coffee is ready.

Based on the requirements above, draw the corresponding flowchart diagram.